| n = 5 | i | # stars |
|---|---|---|
| * | 1 | 1 |
| ** | 2 | 2 |
| *** | 3 | 3 |
| **** | 4 | 4 |
| ***** | 5 | 5 |

i  i stars

```
for( i = n; i >= 1 ; i--)
{
    for( j = 1; j <= i ; j++)
    {
        cout << "*";
    }
    cout << endl;
```

```
}

for(i=1; i<=n; i++)
{
    for(j=1; j<=n; j++)
    {
        if(i==1 or i==n or j==1 or j==n)
        {
            cout << "*";
        }
        else
        {
            cout << " ";
        }
    }
    cout << endl;
}
```

HOLLOW

1) Draw the complete shape

2) Only print *

at the boundary

| num | num % 2 |
| --- | --- |
| $\boxed{1}$ | 1 |
| 2 | 0 |
| 3 | 1 |
| 4 | 0 |
| 5 | 1 |

Switch

$\boxed{\text{Odd / Even}}$

Invert / Reverse

n = 5

```
1   *
2   * *
3   * * *
4   * * * *
5   * * * * *
4   * * * *
3   * * *
2   * *
1   *
```

n = 5

```
----*
---**
--***
-****
*****
```

n

| i | space | star |
|---|---|---|
| 1 | 4 | 1 |
| 2 | 3 | 2 |
| 3 | 2 | 3 |
| 4 | 1 | 4 |
| 5 | 0 | 5 |
| i | n − i | i |

```
for(i=1; i<=n ; i++)
{    //(n-i) space
     for(j=1; j<=(n-i); j++)
     {     cout << " ";
     }

     // i star
```

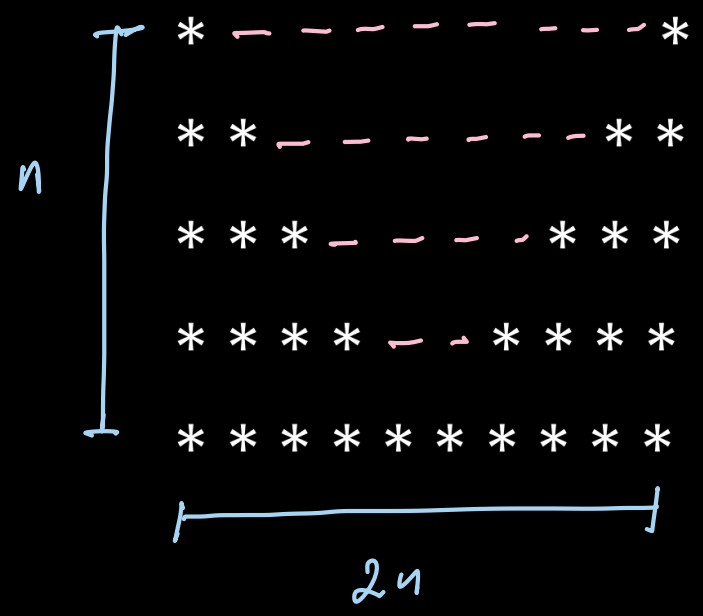
```
for(j=1; j<=i; j++)
{
    cout << "*";
}

cout << endl;
}
```

n = 5

```
- - - - *
- - - * - *
- - * - * - *
- * - * - * - *
* - * - * - * - *
```

| i | spaces | stars |
|---|---|---|
| 1 | 4 | 1 |
| 2 | 3 | 2 |
| 3 | 2 | 3 |
| 4 | 1 | 4 |
| 5 | 0 | 5 |
| $i$ | $n-i$ | $i$ |




| i | stars | spaces | stars |
|---|---|---|---|
| 1 | 1 | 8 | 1 |
| 2 | 2 | 6 | 2 |
| 3 | 3 | 4 | 3 |
| 4 | 4 | 2 | 4 |
| 5 | 5 | 0 | 5 |
| i | i | $\boxed{2n-2i}$ | i |

```
for(i=1 ; i<=n ; i++)
{

    // i stars


    // 2n-2i spaces
```

```
    // i stop

    cout << endl;
}
```